

Project on Healthcare Services

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in partial fulfilment of the requirement for the degree of Bachelor of Science in Computer
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Declaration

We hereby announce that the work is being presented in this project entitled “Healthcare Service” Partial fulfilment of requirements for degree Bachelor of Science in Computer Science and Engineering under Faculty of Engineering and Technology, Eastern University, Bangladesh is an authentic record of our own work done under the supervision of **Muhammad Mahfuz Hasan**. It is also not declared in this report or any part of it has been submitted elsewhere for the award of any degree.

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Abstract

We are currently residing in the era of information technology, where information plays a pivotal role in every facet of modern life. Regrettably, the residents of Bangladesh lag significantly behind the global standards in accessing essential information and various amenities. The absence of a robust healthcare system, coupled with the scarcity of reliable and precise information, compounds the challenges faced by the populace. Consequently, individuals in our country find themselves expending excessive time and undergoing hardships when seeking urgent healthcare services and accurate information. On occasion, this dearth of information leads to irreversible losses for the people. To address these issues, we are developing a website with the primary objective of ensuring that the citizens of Bangladesh have prompt access to healthcare services when needed. We have identified key categories of information that are deemed crucial for the population. This encompasses details about ambulance services, blood banks, and hospital services. Our approach involves not only gathering data on these aspects but also actively engaging with the respective service providers. By doing so, we aim to facilitate seamless access to the necessary data, enabling people to communicate effectively and address their concerns. Furthermore, we are establishing a blood donor system to streamline the process of requesting blood donations and providing accurate information. The overarching Objective of our project is to serve the community by delivering vital information that contributes to the well-being of society and humanity.

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Chapter 1

Introduction

1.1 Introduction

Bangladesh, despite being a compact nation, is home to over 180 million residents. Presently, societal dynamics are evolving as individuals embrace and integrate information technology into their lives. People wield control over virtually everything through technological advancements. However, the challenge persists in acquiring essential information. For instance, when a patient requires blood, reaching out to the most suitable blood donor or obtaining details about the nearest blood bank remains a formidable task, leading to numerous complications.

The "healthcare services information providing system" can be alternatively expressed as the "Healthcare information delivery service," which involves furnishing health care information and support. Our system will consist of three distinct segments: Blood, Ambulance, and Hospital Services.

In the blood department, individuals can solicit blood donors to contribute blood for their patients. Donors will receive notifications via their accounts. Users can also enrol as blood donors, locate the closest blood bank with accurate stock information, and access real-time updates through accounts managed by blood bank authorities. In the ambulance department, ambulance authorities will have dedicated accounts. Regular users can identify nearby ambulance services, submit booking requests via our system, and efficiently secure the needed ambulance based on district and upazila. In the hospital section, individuals can search for their nearest or sub-district hospitals, acquire information about doctors in specific hospitals, and receive details provided by hospital authorities. Specific sub-district information can be obtained in the relevant section.

1.2 Inspiration

People encounter numerous daily challenges and issues, necessitating communication across various domains such as ambulance services, hospitals, blood banks, or blood donors. Unfortunately, obtaining essential healthcare information proves to be a formidable task for many individuals. The process of gathering information and reaching out to service providers is time-consuming. Consequently, there is a significant delay in people accessing crucial information, leading to potential irreparable losses. Our project aims to address these issues comprehensively by consolidating vital information onto a unified platform. This will enable individuals to effortlessly access necessary information and engage in communication with service providers, thereby seeking assistance more efficiently.

1.3 Objectives

The aim of our initiative is to support the residents of Bangladesh by delivering healthcare assistance and accurate information to ensure their well-being. We strive to facilitate easy access to crucial information, enabling direct communication with healthcare service providers such as ambulance services and the Blood Bank department. This streamlined approach aims to mitigate challenges, save time and financial resources, and efficiently address healthcare needs. Additionally, our project aims to foster a community of mutual aid through blood donation. Individuals can appeal for blood donors, and donors can connect with patients in need, potentially saving lives. Our ultimate Objective is to provide round-the-clock access to a comprehensive range of healthcare information through our system, seven days a week, 24/7.

1.4 Project Schedule

Activities	Duration (in Week)	Total week
Idea generation	Week 1, Week 2,	2
Problem identification	Week 3, Week 4, Week 5	3
System Analysis	Week 6, Week 7	2
System Design	Week 8, Week 9	2
Database Design	Week 10, Week 11, Week 12	3
Front-end Design	Week 13, Week 14	2
Backend part and connection with database	Week 15, Week 16, Week 17	3
Test case	Week 18, Week 19	2

Table 1.4 Project Schedule

1.5 Expected Outcome

We anticipate achieving the precise outcomes outlined in our project proposal, aiming to deliver timely healthcare information to the public. Users can expect comprehensive information, ensuring their satisfaction, coupled with a user-friendly interface. Registered individuals and organizations are poised to willingly contribute information and health services, fostering overall public contentment. A seamless process is ensured, guaranteeing that individuals encounter no difficulties in obtaining accurate information on urgent healthcare matters and accessing the necessary services promptly.

Chapter 2

Literature Review

2.1 Introduction

In this chapter, at the beginning we try to give an overall idea and knowledge about the background of the project. Before starting the making of this project, we made a deep study about its background. Which sector will we work with? How do we start this project? After studying the background, we will start the development our project.

2.2 Comparative Studies

We must conduct a comparative analysis to understand whether the envisioned project has been undertaken previously, and if so, the method, platform, and approach employed. A thorough comparative study is crucial for project planning. While certain projects share similarities with our proposed Characteristics, notable distinctions exist among them.

2.3 Feasibility Studies

The feasibility analysis examines the system prerequisites to define our project. The feasibility analysis primarily focuses on the five areas outlined below.

1. Technical Feasibility
2. Operational Feasibility
3. Economic Feasibility
4. Legal Feasibility
5. Schedule Feasibility

2.3.1 Technical Feasibility:

The viability of project development is assessed by examining the technical feasibility, which involves evaluating the current resources encompassing both hardware and software, along with an analysis of the available technology. This process results in a report that determines whether the essential resources and technologies are adequately positioned for the project [1].

Hardware requirements:

1. Processor : Intel Core i5/ AMD A8 or higher
2. RAM: 4 GB or higher
3. Storage: 128 GB HDD/SSD or higher etc.

Software Requirement:

1. Visual Studio(Vs Code)
2. Xampp Server
3. Web Browser

Language and Frameworks Specifications:

1. Html
2. CSS
3. JavaScript
4. PHP
5. jQuery
6. Bootstrap

All the necessary hardware and software are at our disposal, and we possess proficiency in the required language and technologies. Consequently, the system is technically viable.

2.3.2 Operational Feasibility:

Operational feasibility primarily addresses whether the system, once developed and executed, will be effectively utilized. The suggested system is functionally feasible for the following reasons because:

- People can effortlessly gather details regarding blood banks, ambulances, hospitals, and doctors.
- People have the ability to submit blood donation requests to potential blood donors.
- People can effortlessly transmit messages to emergency medical service providers and blood bank authorities.

2.3.3 Economic Feasibility

During the economic viability assessment, an examination of the project's expenditures and returns is conducted. This involves a comprehensive breakdown of the expenses associated with the project's ultimate development, encompassing resources such as hardware and software, design and development outlays, operational expenses, etc. Given that the essential hardware and software components are at our disposal without significant cost considerations, and the expenses related to website hosting and domain acquisition are relatively low, our focus is not profit-oriented but rather geared towards aiding people. Consequently, the project is deemed economically viable.

2.3.4 Legal Feasibility

In legal feasibility assessments, the project is scrutinized in terms of its adherence to legal standards. This involves evaluating obstacles to the legal execution of projects, compliance with data protection and social media laws, project certification, licensing, copyright, etc. We have examined cyber law and copyright law in Bangladesh to ensure that our project complies with all applicable laws in the country. Consequently, the project is deemed legally viable, and we have comprehensively addressed all relevant facets.

2.3.5 Schedule Feasibility

In the Schedule Feasibility Analysis, the primary focus lies in assessing the proposed project's timelines and deadlines. This involves an examination of the anticipated duration for each team to finalize the project [1]. The project submission deadline is set for November 04, 2023. As per our strategic plan, we are confident in meeting this deadline, affirming the feasibility of the project.

2.4 The proposed main Characteristics of the Project

2.4.1 User Characteristics

- Register as a blood donor.
- Send a blood donation request to all blood donors in his/her district.
- Search for the nearest blood bank and send a blood application to the blood bank authority.
- Search for the nearest ambulance and send an ambulance booking request to the ambulance authority.
- Search hospital (it can be searched by name, district, or sub-district)
- View the hospital doctors and filter by their department.
- Contact the Admin

2.4.2 Blood Donors Characteristics

- Login
- Receive request from patient or patient's relatives to donate blood.
- User Profile
- Update Profile

2.4.3 Blood Bank Characteristics

- Login
- Receive blood application from patient or patient relatives
- Update Currently available blood units are in stock.
- User Profile
- Update Profile
- Logout

2.4.4 Ambulance Characteristics

- Login
- Receive ambulance booking requests from users.
- Update current number of available ambulances.
- User Profile
- Update Profile
- Logout

2.4.5 Hospital Characteristics

- Login
- Add and manage doctors.
- User Profile
- Update Profile
- Logout

2.4.6 Admin Characteristics

- Login
- Manage blood donors.
- Manage blood donation requests from users.
- Manage blood bank
- Manage Ambulance
- Manage Hospital
- Receive messages from users.
- Admin profile
- Update Profile
- Logout

2.5 Methodology

We chose the waterfall model for our project. The waterfall model was the first process model. Introduction it is very easy to understand and use. In a waterfall model, each phase must be completed before the next phase begins and there is no overlapping of phases. Following the figure shows the steps of the waterfall model [2].

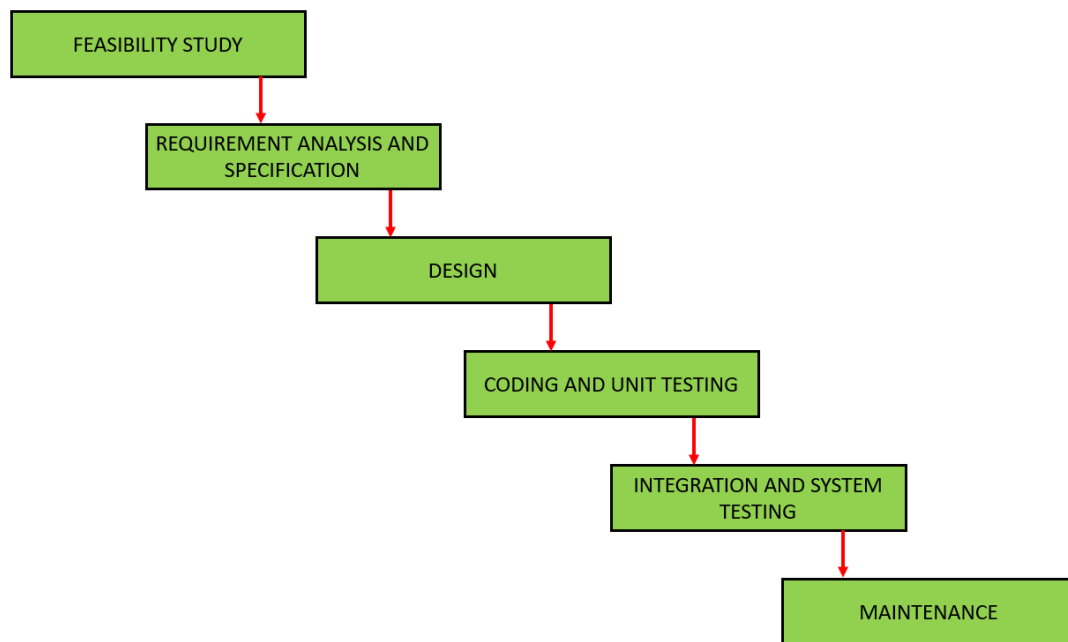


Fig: 2.5 Waterfall model

2.6 Project Management Life Cycle

The project management life cycle outlines overarching procedures for achieving a successful project. Efficient project management helps avoid squandering funds and assets. Throughout the stages of the project management life cycle, we conceive the project idea, establish objectives, strategize for implementation, and steer the project to its conclusion. The project management life cycle comprises five phases [3].

1. Initiation
2. Planning
3. Execution
4. Monitoring And Control
5. Closer



Fig: 2.6 Project Development Life Cycle

2.7 Challenges:

- Engaging individuals in our project
- Engaging Blood Contributors, Blood Centres, Emergency Medical Service Providers, Hospital Administration.

CHAPTER 3

SYSTEM DESIGN AND ANALYSIS

3.1 Introduction

The process of outlining the structure, elements, and data of a system to meet particular requirements and scrutinizing its constituent parts and their interconnections is known as system design and analysis. In this chapter, we will elaborate on the system design and analysis of our project.

3.2 Use Case Modelling and Description:

A utilization-scenario framework is described as a representation utilized to illustrate how individuals engage with a system, aiming to address a particular issue. Utilization-scenario frameworks articulate user Objective, the interplay between the system and the user, and the conduct of the system in fulfilling these essential purposes. These frameworks encompass diverse model components, including participants, utilization scenarios, and connections linking them. Typically, participants are individuals engaged in the designated system based on their functions. A participant could encompass entities such as humans or other external systems. Utilization scenarios delineate how participants employ a system to accomplish a particular Objective. A connection is an additional component of the fundamental framework employed to specify connections between participants and the domains in which they contribute [4].

There are six types of actors using our system:

1. User
2. Blood Donors
3. Blood Bank
4. Ambulance Provider
5. Hospital Authority
6. Admin

3.2.1 Use Case Diagram of the System

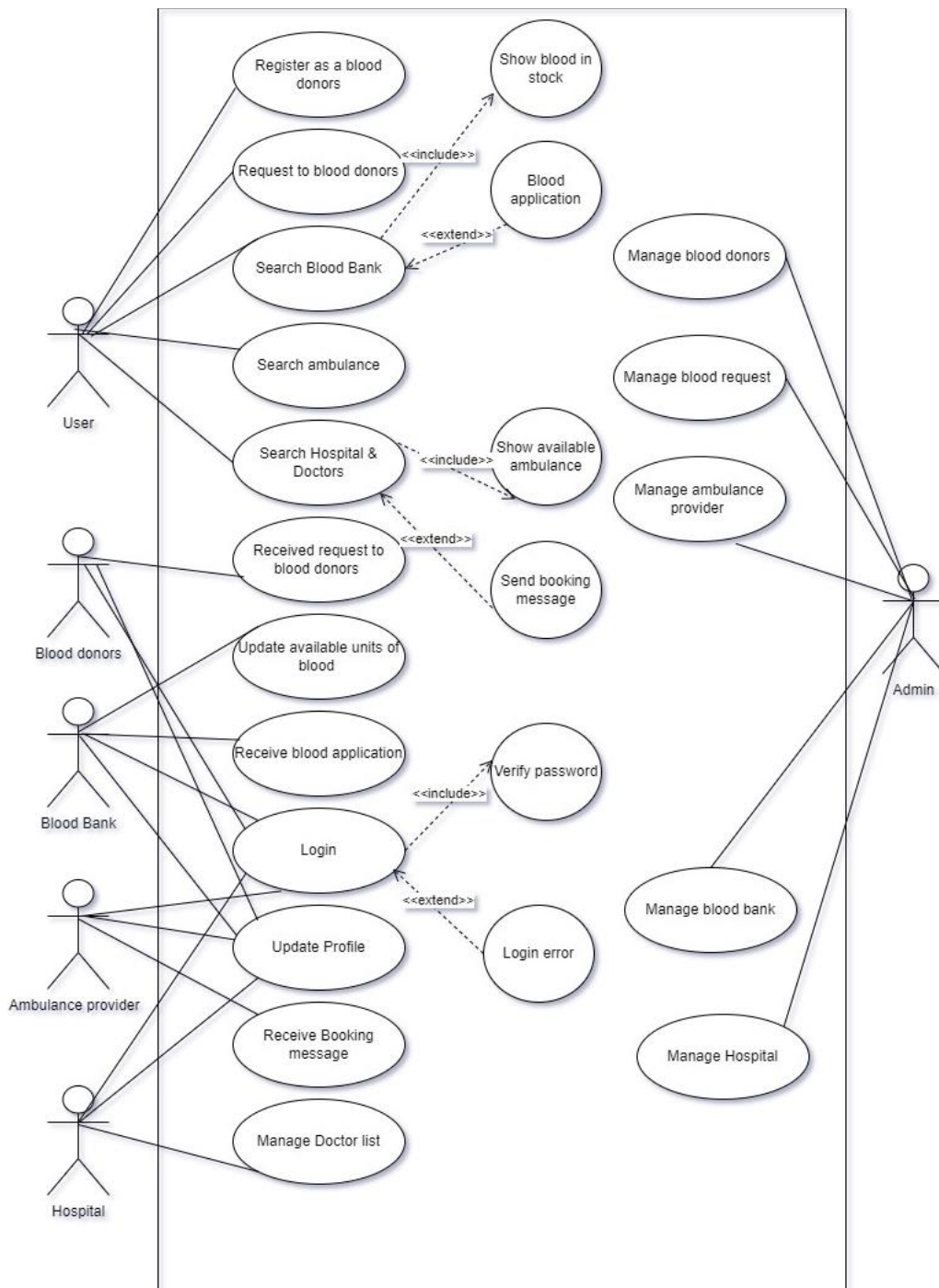


Fig 3.2.1 Use Case Diagram

3.2.2 Use Case Description of User

Objective	Users can seek the necessary details about healthcare services and engage in communication with them.
Condition	Visit our website
Primary Actor	User
Secondary Actor	Admin
Principal Plot	Enlist as a blood donor.
	Search ambulance, blood bank, hospital, doctors.
	Submit blood donation request to the blood bank.
	Compose reservation notifications for emergency medical transport service.
	Contact administrator.

Table 3.2.2 Use case description of User

3.2.3 Use Case Description of Blood Donor

Objective	Blood donors have the ability to log in to the system, where they can view and respond to blood donation requests from users. Additionally, they can modify their profiles as needed.
Condition	Requires a legitimate username and password.
Primary Actor	Blood donors
Secondary Actor	Admin
Principal Plot	Login
	Can see blood donation requests from user
	Update Profile

Table 3.2.3 Use case description of Blood Donor

3.2.4 Use Case Description of Blood Bank

Objective	Blood bank authorities have the capability to access the system and oversee the information pertaining to the blood bank, including details related to blood applications and users.
Condition	Require a legitimate username and password.
Primary Actor	Blood Bank
Secondary Actor	Admin
Principal Plot	Login
	The user is able to view the blood donation request from the user.
	Update the current inventory of available blood units.
	Update Profile

Table 3.2.4 Use case description of Blood Bank

3.2.5 Use Case Description of Ambulance Provider

Objective	Emergency medical services (EMS) authorities have the ability to log in to the system and oversee ambulance details as well as handle booking requests.
Condition	Requires a legitimate username and password.
Primary Actor	Ambulance Provider
Secondary Actor	Admin
Principal Plot	Login
	Can see ambulance booking request from user
	Revise the existing ambulance details.
	Update Profile

Table 3.2.5 Use case description of Ambulance Provider

3.2.6 Use Case Description of Admin

Objective	The administrator has the ability to log into the system and oversee the management of all departments.
Condition	Require a legitimate username and password.
Primary Actor	Admin
Secondary Actor	N/A
Principal Plot	Login
	Coordinate emergency medical services, oversee blood bank operations, mobilize blood donors, facilitate blood donation requests, and manage hospital support services.

Table 3.2.6 Use case description of Admin

3.3 ER Diagram

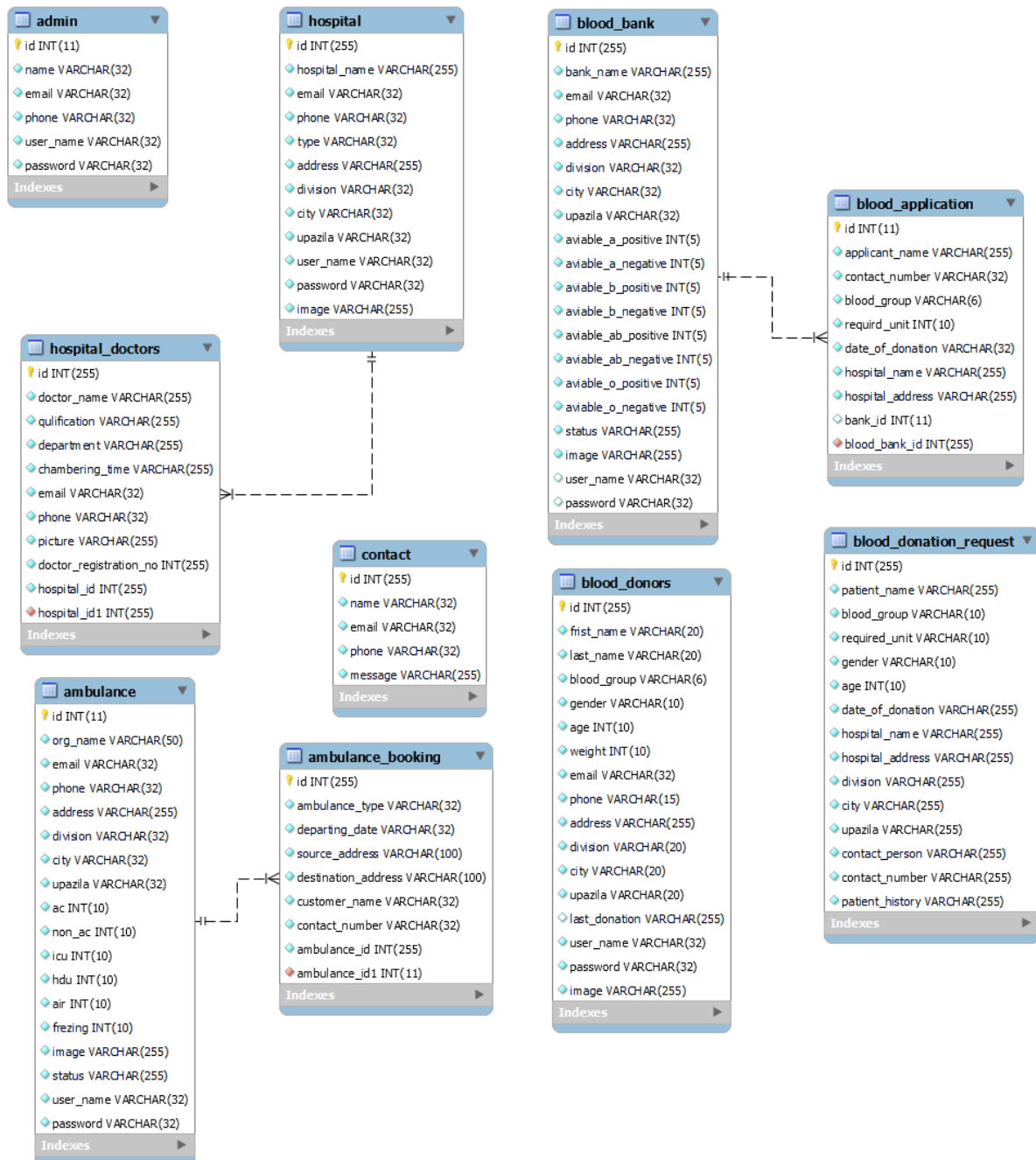


Fig 3.3 ER Diagram

3.4 Data Flow Diagram

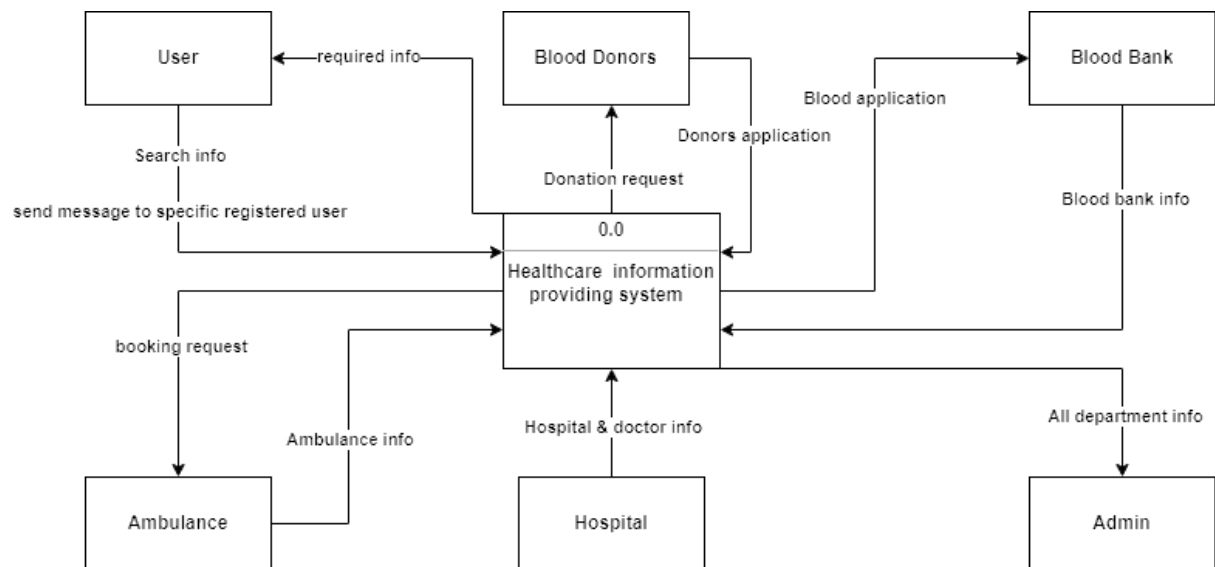


Fig 3.4 Data Flow Diagram

3.5 Activity Diagram

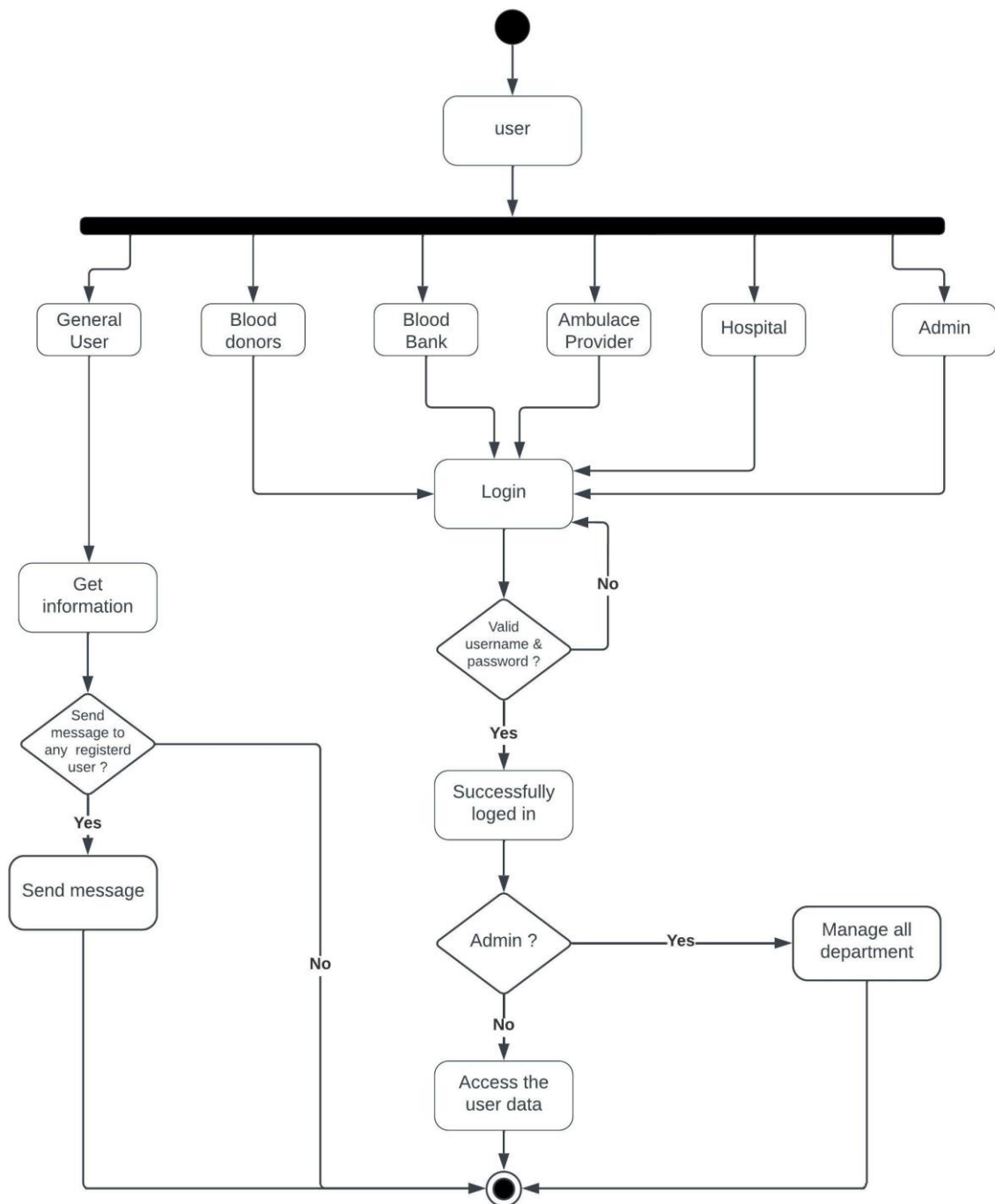


Fig 3.5 Activity Diagram

CHAPTER 4

IMPLEMENTATION AND TESTING

4.1 Introduction

We are adhering to the waterfall methodology in our system development. The execution of the waterfall model commences with input from the system design. The system is initially constructed in compact programs known as units, which are subsequently merged in later phases. Each unit is created and verified for its functionality, a process known as unit testing. Following the testing of each unit, all units developed in the implementation phase are combined into a cohesive system. Post-integration, the complete system undergoes testing to identify any errors or failures.

4.2 Implementation of Database:

We utilize the MYSQL database in our website. SQL stands out as a widely embraced database system in web development, proving particularly well-suited for implementing web applications. It functions as a means to both store and retrieve data for our website. Our project database encompasses a total of 10 tables. The "admin" table captures admin information, the "ambulance" table logs details about ambulance service providers, the "ambulance booking" table records user requests for ambulance bookings, the "Blood Donor" table contains data on blood donors, the "blood_donation_request" table registers blood donation requests from users' donors, the "blood bank" table houses information about blood banks, the "Blood Application" table documents user applications for blood from the blood bank, the "hospital" table contains details about hospitals, the "doctor" table stores information about medical professionals, and the "contact us" table archives user messages directed to the admin.

Table	Action	Rows	Type	Collation	Size	Overhead
admin healthcare service	★ Browse Structure Search Insert Empty Drop	1	InnoDB	utf8mb4_general_ci	16.0 K1B	-
ambulance booking user	★ Browse Structure Search Insert Empty Drop	4	InnoDB	utf8mb4_general_ci	16.0 K1B	-
ambulance service	★ Browse Structure Search Insert Empty Drop	362	InnoDB	utf8mb4_general_ci	112.0 K1B	-
blood bank doner	★ Browse Structure Search Insert Empty Drop	259	InnoDB	utf8mb4_general_ci	96.0 K1B	-
blood contributors	★ Browse Structure Search Insert Empty Drop	1	InnoDB	utf8mb4_general_ci	16.0 K1B	-
blood_application	★ Browse Structure Search Insert Empty Drop	7	InnoDB	utf8mb4_general_ci	16.0 K1B	-
communication	★ Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_general_ci	16.0 K1B	-
hospital authority	★ Browse Structure Search Insert Empty Drop	32	InnoDB	utf8mb4_general_ci	16.0 K1B	-
hospital doctors list	★ Browse Structure Search Insert Empty Drop	79	InnoDB	utf8mb4_general_ci	16.0 K1B	-
plea for blood donation	★ Browse Structure Search Insert Empty Drop	4	InnoDB	utf8mb4_general_ci	16.0 K1B	-
10 tables	Sum	749	InnoDB	utf8mb4_general_ci	336.0 K1B	0 B

Fig 4.2 Database Tables

4.3 Implementation of Front-end Design:

Front-end development, alternatively referred to as client-side development, pertains to the visible and interactive aspects of a website that users directly engage with. Ensuring a polished, appealing, and user-friendly interface in the front-end design is crucial for capturing the attention of clients and sustaining their interest. Our Objective is to streamline the front-end strategy for comprehensibility, thereby optimizing the user's successful utilization of our website.

For our front-end development, we utilized the following languages and frameworks:

4.3.1 Language and framework

HTML:

HTML (Hypertext Markup Language) serves as the fundamental foundation of the Web, acting as the primary cornerstone for defining the significance and framework of web-based content. Our foundational project structure is constructed using HTML.

CSS:

CSS, an acronym for Cascading Style Sheets, dictates the presentation of HTML elements across various mediums such as screens, paper, or other forms of media. It serves as the tool to specify the appearance and arrangement of elements on our web pages, encompassing both design and layout aspects.

Bootstrap:

Bootstrap is a no-cost, open-source framework for front-end development utilized in crafting websites and web applications. Engineered to facilitate the creation of mobile-first websites with adaptive designs, Bootstrap offers a set of syntax for template designs. In our projects, we employ Bootstrap Version 5 (Bootstrap 5), streamlining the process of designing responsive websites without the need for extensive coding.

JavaScript:

JavaScript stands as the most widely adopted programming language globally and serves as the cornerstone of the Web, functioning in tandem with HTML and CSS. Employing JavaScript, we implemented logical solutions to address issues within our project.

JQuery:

We employed jQuery to incorporate certain functionalities into our project, leveraging its capabilities as a swift, compact, versatile, and Characteristics-laden JavaScript library. The primary aim was to streamline the implementation of client-side scripting for HTML, enhancing the interactivity and visual appeal of our website.

4.4 Implementation of back-end development

Back-end development refers to the creation and maintenance of server-side software, addressing aspects of a website that are not visible to the typical user. Back-end developers are responsible for the proper functioning of the website, emphasizing tasks such as managing databases, implementing back-end logic, integrating application programming interfaces (APIs), overseeing architecture, and maintaining servers. PHP has been employed for the back-end development of our project.

PHP:

PHP serves as a versatile backend programming language designed for web applications. It is commonly employed to construct the server-side components of engaging and responsive websites. Our backend development utilizing PHP has been successfully integrated with our MySQL database and is functioning as intended.

4.5 Unit Testing

In the waterfall model, we adhere to the practice of individually developing and testing each unit for its functionality, known as unit testing. This phase entails the examination of every unit or distinct component of the system application and represents the initial stage of functional testing. The primary objective of unit testing is to validate the performance of individual components [5]. We conduct manual testing for each unit within our system, supplemented by the utilization of automated unit testing tools such as "W3 Mark-up Validator Service," "RAKKOTOOLS," and others. Subsequent to the unit test report, we address and rectify all identified bugs. Following the resolution of all bugs, we proceed with the integration of each unit into the overall system.

4.6 Integration Testing

The integration testing phase follows unit testing, focusing on testing a set of interconnected modules collectively. Its objective is to verify the seamless collaboration of modules. Following the completion of integration testing, we have addressed and resolved all identified issues.

4.7 System Testing

System testing is a phase of testing that verifies the functionality and performance of a fully integrated software product. The primary Objective of system testing is to assess the compliance of the entire system with specified end-to-end specifications. This testing process is characterized by black box testing, where the tester inputs data and examines the resultant outputs from the system under examination. Our fully integrated system has undergone comprehensive testing, with each input thoroughly assessed to ensure the generation of the intended outputs. The detailed findings and report are outlined below.

4.7.1 Testing results and report of users' Characteristics

Test case	Test Input/condition	Expected output	Output	Result
Search Blood Bank, Ambulance, Hospital.	Express the Division, District, and Upazila accurately. (Alternatively, search for hospitals by name.)	To present the necessary information	Presented the necessary details.	passed
View doctors from specific hospitals and filter by category	Choose a particular hospital and pick a specific department.	To showcase the doctors associated with a particular department.	Showcased the doctors from the particular department.	passed
Send message/request to blood bank, ambulance or admin	Complete the message/request form with accurate information.	"Message/request successfully dispatched."	"Message/request successfully submitted."	passed

Table 4.7.1 Testing results and report of users' Characteristics

4.7.2 Testing results and report of blood donor Characteristics

Test case	Test input/condition	Expected output	Output	Result
Login	Enter the username and password accurately.	Login successfully	Logged in successfully	passed
Update profile	Required to sign in and update user details.	To present current information	To present current information.	passed
Received a plea for blood donation from a user.	Must log in	To present the blood donation solicitation form to the user.	Present a form for users to request blood donation.	passed
Logout	Must log in	Logout Successfully	Logged out	passed

Table 4.7.2 Testing results and report of blood donor Characteristics

4.7.3 Testing results and report of blood bank Characteristics

Test case	Test input/condition	Expected output	Output	Result
Login	Enter the username and password accurately.	Login successfully	Logged in successfully	passed
Update profile	Required to sign in and update user details.	To present current information.	To showcase current information.	passed
Update blood in stock	You are required to access and update stock details by logging in.	To showcase current blood inventory.	The current inventory of blood is shown.	passed
Received blood Application	Must log in	To showcase the user's blood application.	The user's blood request is shown.	passed
Logout	Must log in	Logout Successfully	Logged out	passed

Table 4.7.3 Testing results and report of blood bank Characteristics

4.7.4 Testing results and report of ambulance Characteristics

Test case	Test input/condition	Expected output	Output	Result
Login	Enter the username and password accurately.	Login successfully	Logged in successfully	passed
Update profile	Required to sign in and update user details.	To present current information.	To showcase current information.	passed
Update number of available ambulance	I have to access the system and adjust the quantity of ambulances that are currently available.	To showcase the quantity of accessible ambulances.	Indicated the quantity of accessible ambulances.	passed
Received booking request	Must log in	To showcase the reservation inquiry initiated by the user.	Present the reservation inquiry submitted by the user.	passed
Logout	Must log in	Logout Successfully	Logged out	passed

Table 4.7.4 Testing results and report of ambulance Characteristics

4.7.5 Testing results and report of admin Characteristics

Test case	Test input/condition	Expected output	Output	Result
Login	Enter the username and password accurately.	Login successfully	Logged in successfully	passed
Manage all department	Must log in	Coordinating blood donors, blood centers, emergency vehicles, and medical facilities.	Effectively supervised every department.	passed
Received messages	Must log in	To showcase messages from users.	Displayed users messages	passed
Update profile	It is required to sign in and update user details.	To showcase the latest details.	Display updated information	Passed
Logout	Must log in	Logout Successfully	Logged out	passed

Table 4.7.5 Testing results and report of admin Characteristics

4.8 Acceptance Testing

At this point, the client will evaluate the system [5]. Since our website hasn't been established, Acceptance testing, a formal testing procedure verifying system adherence to specifications will be performed to:

- Assessing if the system aligns with the acceptance criteria;
- By assessing the fulfillment of user requirements, user satisfaction is ensured.

Currently, the system is undergoing client testing [5]. Since our website is not yet established, we were unable to execute the acceptance test adequately. Moving forward, we plan to conduct

thorough acceptance tests. We presented our project to our supervisor, who provided favourable feedback.

4.9 Snapshot of Front-end

The intended primary functionalities of the project include various pages dedicated to tasks such as information retrieval, information dissemination, user registration, and service provider registration. Visual representations of these pages are presented through selected snapshots.

4.9.1 Home Page

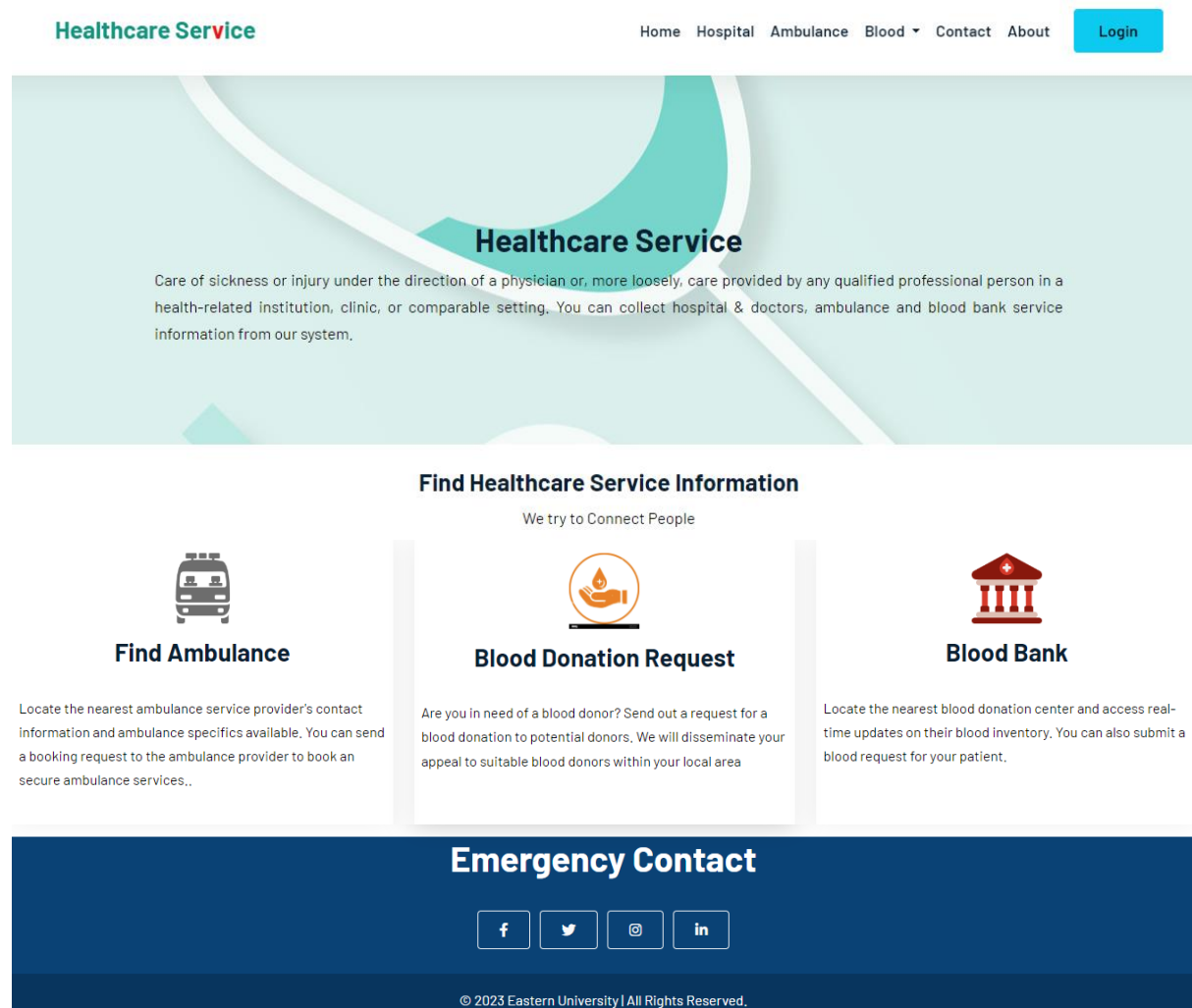
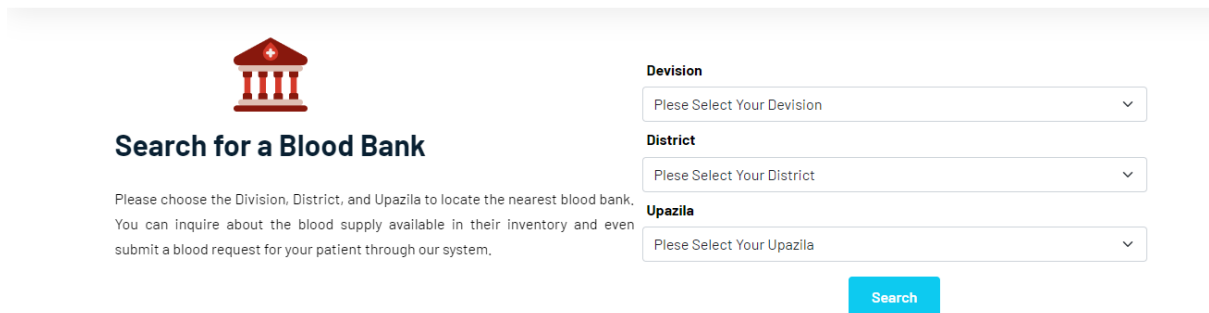


Fig 4.9.1 Home Page

4.9.2 Search Blood Bank



The form is titled "Search for a Blood Bank" and features a red building icon with a white cross. Below the title, a paragraph instructs users to choose a Division, District, and Upazila to find the nearest blood bank and to inquire about blood supply or submit a request. To the right, there are three dropdown menus labeled "Devision", "District", and "Upazila", each with a placeholder text "Plese Select Your [Category]". A blue "Search" button is positioned at the bottom right of the form.

Search for a Blood Bank

Please choose the Division, District, and Upazila to locate the nearest blood bank. You can inquire about the blood supply available in their inventory and even submit a blood request for your patient through our system.

Devision
Plese Select Your Devision

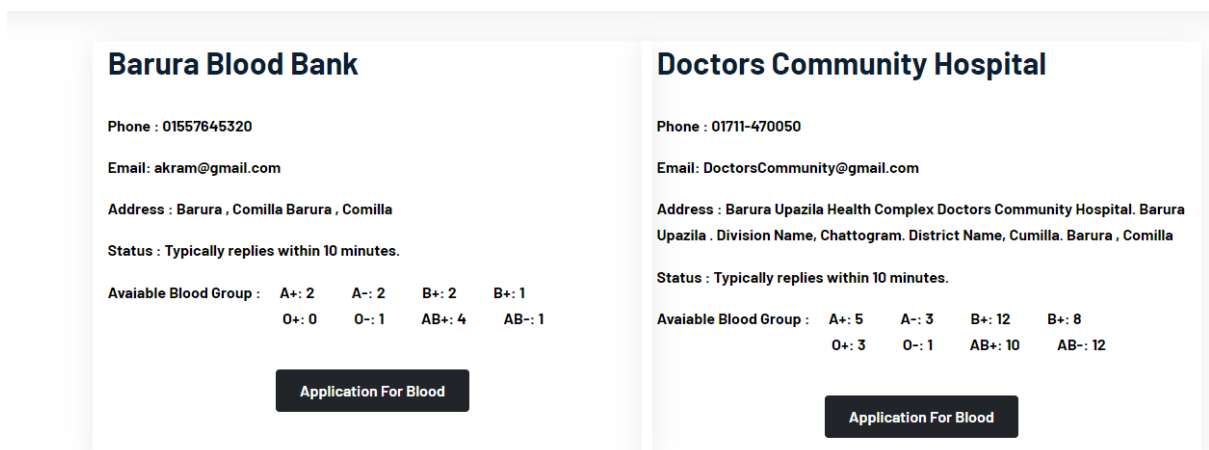
District
Plese Select Your District

Upazila
Plese Select Your Upazila

Search

Fig 4.9.2 Search Blood Bank

4.9.3 Searching Result of Blood Bank



The results are displayed in two columns. The left column is for "Barura Blood Bank" and the right column is for "Doctors Community Hospital". Each column contains contact information (phone and email), address, status, and a table of available blood groups. At the bottom of each column is a dark button labeled "Application For Blood".

Barura Blood Bank

Phone : 01557645320

Email: akram@gmail.com

Address : Barura , Comilla Barura , Comilla

Status : Typically replies within 10 minutes.

Available Blood Group :	A+: 2	A-: 2	B+: 2	B-: 1
	O+: 0	O-: 1	AB+: 4	AB-: 1

Application For Blood

Doctors Community Hospital

Phone : 01711-470050

Email: DoctorsCommunity@gmail.com

Address : Barura Upazila Health Complex Doctors Community Hospital. Barura Upazila . Division Name, Chattogram. District Name, Cumilla. Barura , Comilla

Status : Typically replies within 10 minutes.

Available Blood Group :	A+: 5	A-: 3	B+: 12	B-: 8
	O+: 3	O-: 1	AB+: 10	AB-: 12

Application For Blood

Fig 4.9.3 Searching Result of Blood Bank

4.9.4 Blood Application to Blood Bank

Healthcare Service

Barura Blood Bank

Phone : 01557645320

Email: akram@gmail.com

Address : Barura , Comilla Barura , Comilla

Status : Typically replies within 10 minutes

Avaiable Blood Group : A+: 2 A-: 2
O+: 0 O-: 1

Blood Application

Applicant Name: Md. Noman Jahan

Contact Person Phone: 01831526959

Blood Group: A+ Required Unit: 2

Gender: Male Age: 24

Date of Blood Donation: 23/10/2022

Hospital Name: Doctors Community Hospital

Hospital Address: Barura

Fig 4.9.4 Blood Application to Blood Bank

4.9.5 Searching Hospital

Healthcare Service

Home Hospital Ambulance Blood Contact About Login

Search: barura

Hospital Name	Location	Type	Email	Phone	Doctors
Doctors Community Hospital	Barura Upazila Health Complex Doctors Community Hospital, Barura Upazila , Division Name, Chattogram, District Name, Cumilla.	General	DoctorsCommunity@gmail.com	01711-470050	→

Showing 1 to 1 of 1 entries (filtered from 32 total entries)

Previous 1 Next

Emergency Contact

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Fig 4.9.5 Searching Hospital

4.9.6 Doctors

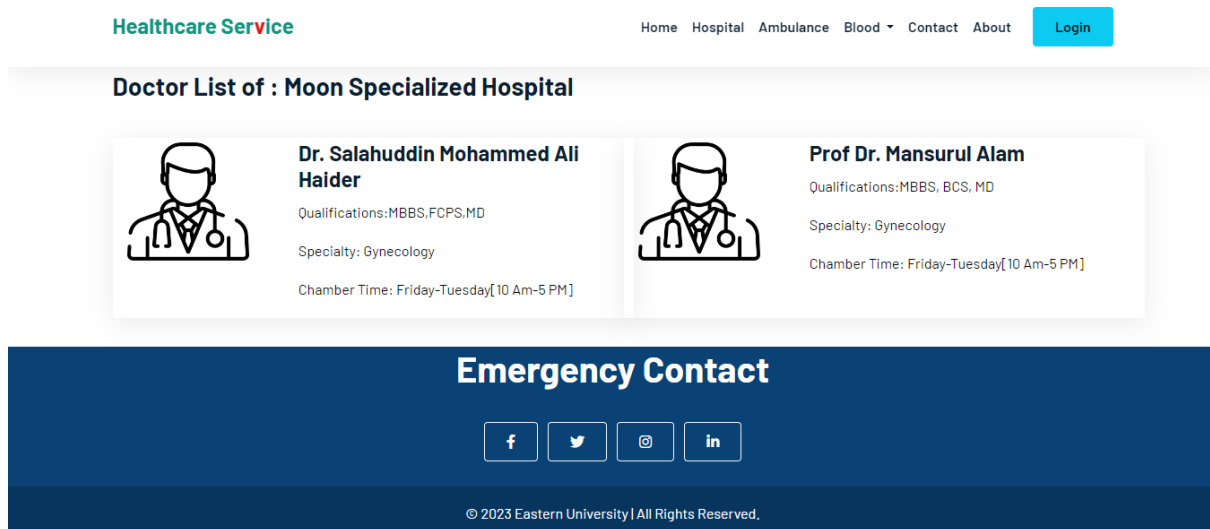


Fig 4.9.6 Doctors

4.9.7 Login Page

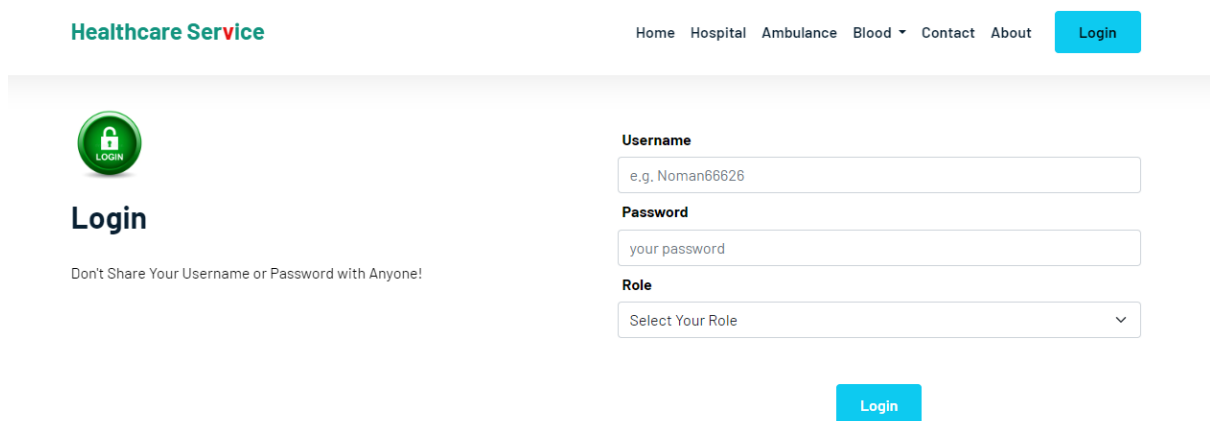


Fig 4.9.7 Login Page

4.9.8 Login as blood donor

4.9.8.1 Blood Donor Profile

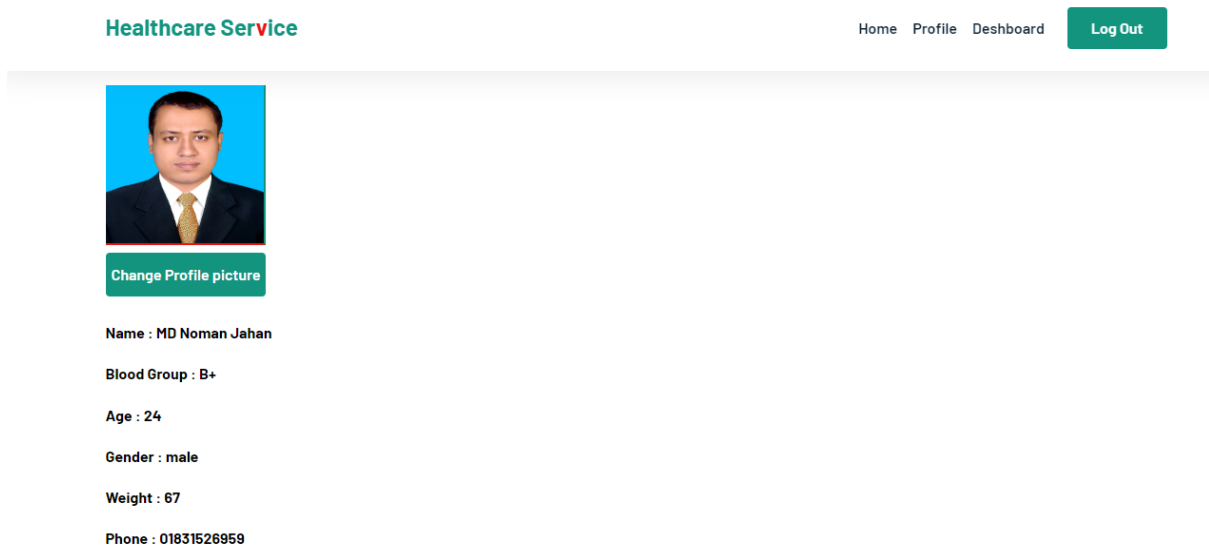


Fig 4.9.8.1 Blood Donor Profile

4.9.8.2 Received blood donation requests from users

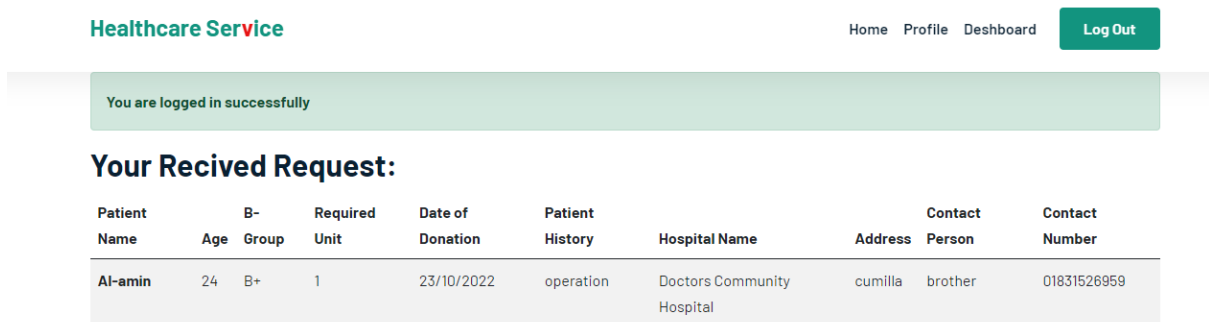


Fig 4.9.8.2 Received blood donation requests from users

CHAPTER 5

CONCLUSION AND FUTURE WORKS

5.1 Conclusion

We have successfully concluded our project, ensuring the proper functioning of all Characteristics and functionalities. Our future endeavors include enhancing and updating the system for modernization. The primary Objective of our project is to furnish the people of Bangladesh with healthcare information, facilitating convenient access to services and enabling them to save time while avoiding complications. We are optimistic that our system will effectively deliver the necessary information, simplifying and securing the lives of individuals. We anticipate that it will play a pivotal role in our mission to digitize the nation, ultimately contributing to the transformation of Bangladesh into a digitally advanced society.

5.2 Future Work

1. Facility for general user account creation.
2. Integrate a live chat/instant messaging functionality facilitating communication between user and blood donors, users and blood bank, or users and ambulance authorities within our platform.
3. Add map facility to find a blood bank, hospital.
4. Provide more information about the hospitals, like facilities, available diagnostic tests, technology, expense, etc.

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